

Secure Rate-Distortion-Perception Trade-Off with Side Information

Gustaf Åhlgren¹ and Onur Günlü^{2, 1}

¹Information Theory and Security Laboratory (ITSL), Linköping University, Sweden

²Lehrstuhl für Nachrichtentechnik, Technische Universität Dortmund, Germany

gustaf.ahlgren@liu.se, onur.guenlue@tu-dortmund.de

Abstract—Secure rate-distortion-perception (RDP) trade-offs are relevant for applications such as semantic compression, where the perceptual quality needs to be maximized. We study a framework for secure RDP over an ideal public communication channel in the presence of an eavesdropper, where the legitimate parties also have access to side information correlated with the source. The exact rate region for the secure RDP trade-off is established when both the encoder and the decoder have access to the side information. We then characterize an inner bound when only the decoder has access to the side information and establish the exact region for a special case. Moreover, we provide an RDP example to illustrate remarkable gains in communication rate due to common randomness, which is not possible to obtain for rate-distortion trade-offs. Our results show that binning-based schemes can achieve high perceptual quality, low distortion, and strong secrecy simultaneously, establishing the information-theoretic limits for next-generation trustworthy semantic compression systems.

I. INTRODUCTION

Modern communication systems increasingly focus on re-constructing the meaningful features of a message rather than exact bit sequences. This shift is driving a significant interest in semantic communications [1], [2], which is of great importance in bandwidth- and latency-limited applications (e.g., autonomous driving and immersive multimedia).

Semantic communications can, from an information-theoretic perspective, be viewed as a remote source coding problem [3]–[5]. We recently introduced the *randomized distributed function computation* (RDFC) framework [6], which incorporates controlled encoder randomization to generate outputs following a target distribution, while preserving utility. RDFC applies to differential privacy mechanisms [6]–[9], neural compression with generative models [10], [11], and federated learning with side information [12]. The RDFC framework achieves strong performance guarantees by leveraging strong coordination results [13], even when common randomness is minimal or absent [6], [14]–[16]. Moreover, leveraging its relevance to image compression, we focus on the *rate-distortion-perception* (RDP) problem [17, Section 17.4.2], [18]–[23]. The RDP problem sets out to both minimize the expected distortion between the reconstructed image and the original image, while simultaneously constraining the distribution of the reconstruction to be close to the source distribution. The second constraint preserves perceptual realism by ensuring that the generated outputs follow

the statistics of the original image [24], [25]. This notion abstracts, e.g., deep learning-based generative discriminators [26]–[28].

Moreover, security and privacy are of great importance in practical implementations where eavesdroppers may observe the public outputs of the encoders. Physical layer security (PLS) techniques are attractive for protection against such attacks [29], [30]. For instance, the methods designed for joint source-channel coding for secure image compression [31] align with these RDP models, which aim to limit or eliminate the information gained by an adversary. Furthermore, in [32] the influence of side information, which was studied for the rate-distortion (RD) function, has recently been extended to the RDP problem under the perceptual realism constraint described above [33]–[35]. We study an RDFC application that combines the RDP problem with side information and PLS by imposing a strong secrecy constraint on the generated outputs, i.e., secure RDP with side information. We study two versions of this problem. First, we assume that the side information is available at both the encoder and the decoder. Second, the side information is only available at the decoder. We characterize the secure RDP region with side information when both the encoder and the decoder have access to the side information. We further derive an inner bound for the secure RDP problem with side information available only at the decoder, shown to be tight for a special case. Finally, we provide a simplified example to illustrate substantial communication rate reductions due to available common randomness.

Notation: Denote the set of integers $\{a, a+1, \dots, b\}$ as $[a : b]$. $X \sim \text{Unif}[a : b]$ represents a random variable X that is uniformly distributed on the interval between a and b . The total variation distance between two probability distributions is given by $\|P_Y - P_X\|_{\text{TV}} \triangleq \frac{1}{2} \sum_{b \in \mathcal{B}} |P_Y(b) - P_X(b)|$. Denote the set of indices $\{1, 2, \dots, i-1, i+1, \dots, n\}$ as $\{n\} \setminus i$.

II. SYSTEM MODEL

Let $X^n \in \mathcal{X}^n$ be an independent and identically distributed (i.i.d) sequence distributed according to Q_X^n . X^n is observed by an encoder (Enc) who also has access to common randomness $C \in [1 : 2^{nR_0}]$ shared with the decoder (Dec). The common randomness is independent of the source sequence X^n and uniformly distributed over $[1 : 2^{nR_0}]$ and available

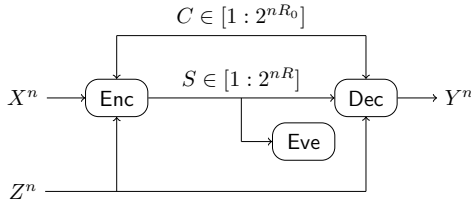


Fig. 1. System model for the secure RDP problem with side information at both the encoder and the decoder and with noiseless transmissions.

only to the encoder and decoder. There exists a memoryless side information sequence $Z^n \in \mathcal{Z}^n$ that is correlated with the source sequence X^n according to Q_{XZ}^n and independent of the common randomness. We consider two different schemes with the side information. In the first case, both the encoder and the decoder have access to the side information, and in the second case, only the decoder has access to the side information. The two system models are depicted in Fig. 1 and Fig. 2, respectively. In both cases, the side information is kept secret from the Eve.

The encoder produces an index $S \in [1 : 2^{nR}]$, which is a function of X^n , C , and the side information Z^n (if available) that is sent over a noiseless channel to the decoder. The decoder produces an output $Y^n \in \mathcal{Y}^n$ such that (i) the per-letter distortion between X^n and Y^n is minimized, (ii) the distribution of the output sequence P_{Y^n} is close in total variation distance to Q_X^n , and (iii) the index S , observed also by the Eve, is statistically independent of the output sequence Y^n , i.e., strong secrecy between S and Y^n . This secrecy constraint is relevant in the context of generative artificial intelligence, where we are interested in keeping the generated output secret at the commodity [16].

We define the achievable region in Definition 1 below.

Definition 1: A secure RDP tuple (R, R_0, D) is achievable with side information $Z^n \in \mathcal{Z}^n$, correlated with the source sequence X^n according to Q_{XZ}^n , if, for any $\epsilon > 0$ there exists $n \geq 1$, an encoder, and a decoder such that

$$\|P_{Y^n} - Q_X^n\|_{TV} \leq \epsilon, \quad (\text{realism}) \quad (1)$$

$$\mathbb{E}[d(X^n, Y^n)] \leq D + \epsilon, \quad (\text{distortion}) \quad (2)$$

$$I(Y^n; S) \leq \epsilon \quad (\text{strong secrecy}) \quad (3)$$

with a per-letter distortion metric $d(x^n, y^n) = \frac{1}{n} \sum_{i=1}^n d(x_i, y_i)$. The rate region for the secure RDP with side information $\mathcal{R}_{\text{SI}}$ is the closure of the set of all achievable tuples. $\diamond$

Definition 2: Two sequences of random variables A^n and B^n are jointly i.i.d if and only if we have

$$P_{A^n B^n} = P_{AB}^n. \quad (4)$$

$\diamond$

We next provide the rate region for secure RDP with side information.

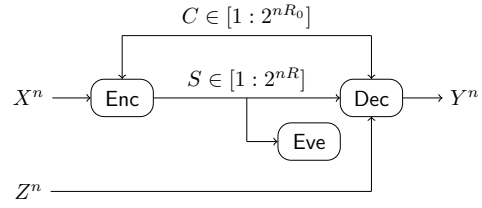


Fig. 2. System model for the secure RDP problem with side information at the decoder only and transmission occurring over a noiseless channel.

III. RATE REGIONS FOR RDP WITH SIDE INFORMATION

We characterize the rate region $\mathcal{R}_{\text{SI,ED}}$ for secure RDP with side information at both the encoder and the decoder in Theorem 1. Proof sketch of Theorem 1 is given in Section IV-A.

Theorem 1: The region $\mathcal{R}_{\text{SI,ED}}$ is the union over all joint probability distributions $P_{XZUY} = Q_{XZ} P_{UY|XZ}$ of the set of all (R, R_0, D) tuples that satisfy

$$R \geq I(U; X|Z), \quad (5)$$

$$R_0 \geq I(U; Y) - I(U; Z), \quad (6)$$

$$R + R_0 \geq I(U; Y|Z) - H(Z|Y), \quad (7)$$

$$D \geq \mathbb{E}[d(X, Y)] \quad (8)$$

such that $X - (U, Z) - Y$ forms a Markov chain and the cardinality $|\mathcal{U}|$ of U can be limited to $|\mathcal{X}|^2 |\mathcal{Z}| + 2$.

Next, we characterize the rate region $\mathcal{R}_{\text{SI,D}}$ in Theorem 2. Proof sketch of Theorem 2 is given in Section V.

Theorem 2: The region $\mathcal{R}_{\text{SI,D}}$ includes the union over all joint probability distributions $P_{XZUY} = Q_{XZ} P_{UY|XZ}$ of the set of all (R, R_0, D) tuples that satisfy

$$R \geq I(U; X|Z), \quad (9)$$

$$R_0 \geq I(U; Y) - I(U; Z), \quad (10)$$

$$R + R_0 \geq I(U; Y|Z), \quad (11)$$

$$D \geq \mathbb{E}[d(X, Y)] \quad (12)$$

such that $X - (U, Z) - Y$ and $Z - X - U$ form Markov chains.

Next, we establish the exact rate region $\mathcal{R}_{\text{SI,D}}$ for the special case where the joint distribution of the output of the decoder and the side information is jointly i.i.d. The proof of Corollary 1 is given in Section VI.

Corollary 1: The rate region $\mathcal{R}_{\text{SI,D}}$ for when the decoder output and the side information are jointly i.i.d. is the union over all joint probability distributions $P_{XZUY} = Q_{XZY} P_{U|XZY}$ of the set of all (R, R_0, D) tuples that satisfy

$$R \geq I(U; X|Z), \quad (13)$$

$$R_0 \geq I(U; Y) - I(U; Z), \quad (14)$$

$$R + R_0 \geq I(U; Y|Z), \quad (15)$$

$$D \geq \mathbb{E}[d(X, Y)] \quad (16)$$

such that $X - (U, Z) - Y$ and $Z - X - U$ form Markov chains and the cardinality $|\mathcal{U}|$ of U can be limited to $|\mathcal{X}|^2 |\mathcal{Z}| + 2$.

Remark: Corollary 1 differs from the region derived in [33, Theorems 17 and 18] where the jointly i.i.d nature of Y^n and Z^n is explicitly imposed.

IV. PROOF OF THEOREM 1

We next provide the achievability and converse proof for Theorem 1.

A. Achievability

Proof sketch: The achievability proof applies the method of output statistics of random binning (OSRB) [36].

Fix a probability distribution P_{XZUY} such that $\mathbb{E}[d(X, Y)] \leq D + \epsilon_n$, where $\epsilon_n > 0$ and $\epsilon_n \rightarrow 0$ when $n \rightarrow \infty$. Generate an auxiliary random variable sequence U^n in an i.i.d. fashion such that we have a joint distribution $P_{XZU} = Q_{XZ}P_{U|XZ}$. Let F be another random variable drawn uniformly at random according to $P_F = \text{Unif}[1 : 2^{n\tilde{R}}]$, which serves as additional common randomness in the problem. The encoder and decoder have access to this additional common randomness F , which represents the public choice of the random codebook. The key idea of OSRB proofs is to assign the bin indices C , S , and F (for this system) in two ways and show that their analyses provide asymptotically equivalent results.

The first way applies *random binning* that assigns a random bin to each u^n , i.e., $C = \varphi_1(u^n)$ $\varphi_1 : \mathcal{U}^n \rightarrow [1 : 2^{nR_0}]$. Similarly random binning setps assign an index $S \in [1 : 2^{n\tilde{R}}]$ and $F \in [1 : 2^{n\tilde{R}}]$ for each u^n uniformly at random. These indices are shared with the receiver such that it can recover U^n from (C, F, S, Z^n) using a decoder $P_{\hat{U}^n|CSFZ^n}^{\text{RB}}$ with the output $\hat{U}^n$, which is then mapped to Y^n with $P_{Y^n|\hat{U}^n}^{\text{RB}}$.

The second way applies *random coding*, where the common randomness C and the public random encoder-decoder choice F are generated independently and uniformly at random according to P_C and P_F respectively. The auxiliary random variable U^n is generated from (X^n, Z^n, C, F) . The index S is selected according to $P_{S|U^n}^{\text{RB}} \sim \text{Unif}[1 : 2^{n\tilde{R}}]$, which is the same way as in the random binning. The decoder that recovers U^n , with the output $\hat{U}^n$, and the mapping from the decoder output to Y^n is the same for the random coding as in the random binning.

The induced distributions in the random binning P^{RB} and the random coding P^{RC} are

$$\begin{aligned} P^{\text{RB}} &= Q_{XZ}^n P_{U|XZ}^n P_C P_{F|U^n} P_{S|U^n}^{\text{RB}} P_{\hat{U}^n|CSFZ^n}^{\text{RB}} P_{Y^n|\hat{U}^n}^{\text{RB}} \\ &= P_{X^n Z^n C F} P_{U^n|CFX^n Z^n} P_{S|U^n}^{\text{RB}} P_{\hat{U}^n|CSFZ^n}^{\text{RB}} P_{Y^n|\hat{U}^n}^{\text{RB}}, \end{aligned} \quad (17)$$

$$P^{\text{RC}} = Q_{XZ}^n P_C P_F P_{U^n|CFX^n Z^n} P_{S|U^n}^{\text{RB}} P_{\hat{U}^n|CSFZ^n}^{\text{RB}} P_{Y^n|\hat{U}^n}^{\text{RB}}. \quad (18)$$

With similar arguments as in [37, Section IV.A] and [16, Section IV.A], which can be applied since both the random coding and the random binning use the same encoding and decoding functions, the total variation distance between P^{RB} and P^{RC} can be upper bounded as

$$\|P^{\text{RB}} - P^{\text{RC}}\|_{\text{TV}} \leq \|P_{X^n Z^n C F} - Q_{XZ}^n P_C P_F\|_{\text{TV}}. \quad (19)$$

The total variation distance in (19) tend to zero when $n \rightarrow \infty$ if we impose the following rate constraints.

The index C and the public index F are almost independent of (X^n, Z^n) and jointly uniformly distributed if we impose [36, Theorem 1]

$$\tilde{R} + R_0 < H(U|XZ). \quad (20)$$

The public indices S and F are almost independent of Y^n , satisfying the secrecy constraint (3) and jointly uniformly distributed, if we have [36, Theorem 1]

$$R + \tilde{R} < H(U|Y). \quad (21)$$

Lastly, fix the decoder $P_{\hat{U}^n|CSFZ^n}^{\text{RB}}$ to be a Slepian-Wolf (SW) decoder $P_{\hat{U}^n|CSFZ^n}^{\text{SW}}$ which can reliably estimate U^n from (C, S, F, Z^n) such that the sequence estimation error goes to zero when $n \rightarrow \infty$ if we impose [36, Lemma 1]

$$R + \tilde{R} + R_0 > H(U|Z). \quad (22)$$

Applying Fourier-Motzkin elimination [38, Appendix D] to the rates given by (20) - (22), we obtain:

$$R = I(U; X|Z) + \delta, \quad (23)$$

$$R_0 = I(U; Y) - I(U; Z) + \delta \quad (24)$$

where $\delta > 0$ and $\delta \rightarrow 0$ when $n \rightarrow \infty$. Moreover, similar to [33, Section IV] we apply [13, Corollary VII.5] to the system in Fig. 1 which yields the following rates, R and $R + R_0$, for the induced codebook to satisfy the realism constraint (1)

$$R \geq I(U; X|Z), \quad (25)$$

$$R + R_0 \geq I(U; Y|Z) - H(Z|Y). \quad (26)$$

By (19) and (20), it follows that the distributions induced by the random binning and random coding schemes coincide asymptotically. Property 3) in [36, Lemma 4] ensures the existence of an instance of $F=f$ such that the total variation distance between P^{RB} and P^{RC} tends to zero when $n \rightarrow \infty$ conditioned on this instance of $F=f$. This property ensures that we can remove the common randomness introduced at the beginning of the proof. Furthermore, since all sequence tuples (x^n, z^n, u^n, y^n) are in the jointly typical set with high probability, the typical average lemma [38, pp. 26] ensures that the distortion constraint (2) is satisfied. ■

B. Converse

Proof outline: Suppose there exists a tuple (R, R_0, D) such that (1) - (3) are satisfied for some $\epsilon > 0$ and $n \geq 1$ in the system shown in Fig. 1. Let $T \sim [1 : n]$ be a time-sharing random variable independent of all system model random variables and define the auxiliary random variable $U_i = (S, C, Z_{\{n\} \setminus i})$ such that $X_i - (U_i, Z_i) - Y_i$ forms a Markov chain. The converse bounds on R and $R + R_0$ follow from [33, Eq. (23)-(27)] with $[X]^n = X^n$, $Z^n = Z^n$, $M = S$, $J = C$, and $V_T = (S, C, Z_{\{n\} \setminus T})$. For the converse bound on R_0 , we have:

$$\begin{aligned} nR_0 &\geq H(C) - H(C|SY^n Z^n) + H(C|SY^n Z^n) \\ &\stackrel{(a)}{=} H(C) - H(Y^n|SCZ^n) - H(C|SZ^n) + H(Y^n|SZ^n) \\ &\quad + H(C|SY^n Z^n) \end{aligned}$$

$$\begin{aligned}
& \stackrel{(b)}{=} H(C) - H(Y^n|SCZ^n) - H(C|SZ^n) + H(Y^n|SZ^n) \\
& \quad + H(Z^n|SC) + H(C|S) + H(Y^n|SCZ^n) \\
& \quad - H(Z^nY^n|S) \\
& \stackrel{(c)}{=} H(C) - H(Y^n|SCZ^n) - H(C|SZ^n) + H(Y^n|S) \\
& \quad + H(Z^n|SY^n) - H(Z^n|S) + H(Z^n|SC) \\
& \quad + H(C|S) + H(Y^n|SCZ^n) - H(Z^nY^n|S) \\
& \stackrel{(d)}{=} H(Y^n|S) - H(Y^n|SCZ^n) - H(Z^n|S) + H(Z^n|SC) \\
& \quad + H(C) - H(C|SZ^n) + H(C|S) + H(Z^n|SC) \\
& \quad + H(Y^n|SCZ^n) - H(Y^nZ^n|S) \\
& \stackrel{(e)}{=} H(Y^n) - H(Y^n|SCZ^n) - H(Z^n|S) + H(Z^n|SC) \\
& \quad + H(C) - H(C|SZ^n) + H(C|S) + H(Z^n|SC) \\
& \quad + H(Y^n|SCZ^n) - H(Y^nZ^n|S) - g(\epsilon) \\
& \stackrel{(f)}{=} H(Y^n) - H(Y^n|SCZ^n) - H(Z^n|S) + H(Z^n|SC) \\
& \quad + H(C) - H(C|SZ^n) + H(Y^nZ^nC|S) \\
& \quad - H(Y^nZ^n|S) - g(\epsilon) \\
& \geq H(Y^n) - H(Y^n|SCZ^n) - H(Z^n) + H(Z^n|SC) \\
& \quad + H(C) - H(C|SZ^n) + H(Y^nZ^nC|S) \\
& \quad - H(Y^nZ^n|S) - g(\epsilon) \\
& \geq H(Y^n) - H(Y^n|SCZ^n) - H(Z^n) + H(Z^n|SC) \\
& \quad - g(\epsilon) \\
& \stackrel{(g)}{\geq} \sum_{i=1}^n [H(Y_i) - H(Y_i|SCZ^nY^{i-1}) \\
& \quad - (H(Z_i) - H(Z_i|SCZ^{i-1})) - f(\epsilon)] - g(\epsilon) \\
& \geq \sum_{i=1}^n [H(Y_i) - H(Y_i|SCZ_{\{n\}\setminus i}) \\
& \quad - (H(Z_i) - H(Z_i|SCZ_{\{n\}\setminus i})) - f(\epsilon)] - g(\epsilon) \\
& = n[H(Y_T|T) - H(Y_T|SCZ_{\{n\}\setminus T}T) - (H(Z_T|T) \\
& \quad - H(Z_T|SCZ_{\{n\}\setminus T}T))] - nf(\epsilon) - g(\epsilon) \\
& \stackrel{(h)}{=} n[I(U; Y) - I(U; Z)] - nf(\epsilon) - g(\epsilon) \tag{27}
\end{aligned}$$

where (a), (b), and (c) follow from chain rule expansions (28)-(30) given below to obtain all the terms other than $H(C)$

$$\begin{aligned}
H(C|SY^nZ^n) &= H(Y^n|SCZ^n) + H(C|SZ^n) \\
&\quad - H(Y^n|SZ^n), \tag{28}
\end{aligned}$$

$$\begin{aligned}
H(C|SZ^nY^n) &= H(Z^n|SC) + H(C|S) + H(Y^n|SCZ^n) \\
&\quad - H(Z^nY^n|S), \tag{29}
\end{aligned}$$

$$H(Y^n|SZ^n) = H(Y^n|S) + H(Z^n|SY^n) - H(Z^n|S), \tag{30}$$

(d) follows from reordering the terms, (e) follows from the secrecy constraint (3) where $g(\epsilon)$ is a function that tends to zero as $n \rightarrow \infty$, (f) follows from the chain rule, (g) follows from the i.i.d nature of Z^n , and because Y^n is almost i.i.d due to the realism constraint (1) where $f(\epsilon)$ is a function that tends to zero as ϵ goes to zero, and (h) follows from identifying the random variables $Y = (Y_T, T)$, $Z = (Z_T, T)$ and the auxiliary random variable $U = (U_T, T)$.

We have

$$\begin{aligned}
D + \epsilon &\stackrel{(a)}{\geq} \mathbb{E}[d(X^n, Y^n)] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[d(X_i, Y_i)] \\
&\stackrel{(b)}{\geq} \mathbb{E}[d(X_T, Y_T)|T] + \epsilon' \stackrel{(c)}{=} \mathbb{E}[d(X, Y)] + \epsilon' \tag{31}
\end{aligned}$$

where (a) follows by (2), (b) follows since Y^n is almost i.i.d for an ϵ' such that $\epsilon' \rightarrow 0$ when $n \rightarrow \infty$ and (c) follows by identifying $X = X_T$ and $Y = Y_T$.

Finally, using the support lemma [5, Lemma 15.4], the probability distribution P_{XZY} can be preserved using $|\mathcal{X}||\mathcal{Z}||\mathcal{Y}| - 1 = |\mathcal{X}|^2|\mathcal{Z}| - 1$ continuous real-valued functions. In addition to that, one should preserve three more terms in Theorem 1, and thus imposing $|\mathcal{X}|^2|\mathcal{Z}| + 2$ suffices. ■

V. PROOF OF THEOREM 2

Next, we provide the achievability proof for Theorem 2, which is proved by using the OSRB method.

Proof sketch: Fix a distribution P_{XZUY} such that $\mathbb{E}[d(X, Y)] \leq D + \epsilon_n$, where $\epsilon_n > 0$ such that $\epsilon_n \rightarrow 0$ when $n \rightarrow \infty$. Generate an auxiliary random variable U^n in an i.i.d manner such that we have a joint distribution of (X^n, Z^n, U^n, Y^n) marginalizing as $P_{XZU} = Q_{XZ}P_{U|X}$. Let F be a random variable distributed according to $P_F = \text{Unif}[1 : 2^{n\tilde{R}}]$, which serves as additional common randomness in the problem. Suppose that the encoder and the decoder have access to this additional common randomness that represents the public choice of the random encoder and decoder.

Next, we randomly assign the indices C , S , and F using *random binning* and *random coding*, as in Section IV-A. Similarly, both the random binning and the random coding select the index S according to $P_{S|U^n}^{\text{RB}} \sim \text{Unif}[1 : 2^{nR}]$ and have a receiver that recovers U^n from (C, F, S, Z^n) using a decoder $P_{\hat{U}^n|CSFZ^n}^{\text{RB}}$ with the output $\hat{U}^n$. The decoder output $\hat{U}^n$ is then mapped to Y^n with $P_{Y^n|\hat{U}^n}^{\text{RB}}$. Importantly, in this case, the auxiliary random variable sequence U^n is generated from (C, F, X^n) in the random coding.

The induced distributions in the random binning P^{RB} and the random coding P^{RC} are

$$\begin{aligned}
P^{\text{RB}} &= Q_{XZ}^n P_{U|X}^n P_C P_{U^n|F} P_{F|U^n} P_{S|U^n}^{\text{RB}} P_{\hat{U}^n|CSFZ^n}^{\text{RB}} P_{Y^n|\hat{U}^n} \\
&= P_{X^nZ^nCF} P_{U^n|CFX^n} P_{S|U^n}^{\text{RB}} P_{\hat{U}^n|CFZ^n}^{\text{RB}} P_{Y^n|\hat{U}^n}^{\text{RB}}, \tag{32}
\end{aligned}$$

$$P^{\text{RC}} = Q_{XZ}^n P_C P_F P_{U^n|CFX^n} P_{S|U^n}^{\text{RB}} P_{\hat{U}^n|CFZ^n}^{\text{RB}} P_{Y^n|\hat{U}^n}^{\text{RB}}. \tag{33}$$

With the same reasoning as in Section IV-A, the total variation distance between the induced distributions from the random binning P^{RB} and the random coding P^{RC} can be upper bounded as follows

$$\|P^{\text{RB}} - P^{\text{RC}}\|_{\text{TV}} \leq \|P_{X^nZ^nCF} - Q_{XZ}^n P_C P_F\|_{\text{TV}}. \tag{34}$$

The availability of side information only affects the strong coordination part of the proof. Therefore, following the same reasoning as in Section IV-A, (20) is imposed to ensure

that the total variation distance in (34) goes to zero, (21) is imposed to satisfy the secrecy constraint (3), and the decoder is fixed as a SW decoder which can recover U^n from (C, F, S, Z^n) if we impose (22). The Fourier-Motzkin elimination applied to the rates imposed above yields the following rates

$$R = I(U; X|Z) + \delta, \quad (35)$$

$$R_0 = I(U; Y) - I(U; Z) + \delta \quad (36)$$

where $\delta \rightarrow 0$ when $n \rightarrow \infty$.

Moreover, [13, Corollary VII.6] applied in the system in Fig. 2 yields the following sum-rate for achieving the realism constraint (1) with local channel synthesis in the decoder using the side information Z^n

$$R + R_0 \geq I(U; Y|Z). \quad (37)$$

By combining (34) with the rate imposed in (20), the probability distributions induced by the random binning and random coding constructions are asymptotically identical. Property 3 of [36, Lemma 4] guarantees the existence of an instance $F = f$ such that the total variation distance between P^{RB} and P^{RC} , conditioned on this instance, vanishes as $n \rightarrow \infty$. Furthermore, since the sequence tuple (x^n, z^n, u^n, y^n) lies in the jointly typical set with high probability, the typical average lemma [38, pp. 26] implies that the distortion requirement in (2) is satisfied. ■

VI. PROOF OF COROLLARY 1

The achievability proof for Corollary 1 follows from Theorem 2. The converse proof for Corollary 1 is given below.

Proof outline: Suppose there exists a tuple (R, R_0, D) such that (1)-(3) are satisfied for some $\epsilon > 0$ and $n \geq 1$ in the system shown in Fig. 2, and where the output of the decoder Y^n and the side information Z^n is jointly i.i.d according to Definition 2. Let $T \sim [1 : n]$ be a time-sharing random variable independent of all other random variables and define the auxiliary random variable $U_i = (S, C, Z_{\{n\} \setminus i})$ such that $X_i - (U_i, Z_i) - Y_i$ and $Z_i - X_i - U_i$ form Markov chains.

For the converse bound on R we use [33, Eqs. (23)-(25)] with $[X]^n = X^n$, $Z^n = Z^n$, $M = S$, $J = C$, and $(V, T) = (U_T, (S, C, Z_{\{n\} \setminus T}))$. The converse bound on the sum-rate $R + R_0$ follows from [33, Eq. (84)] with the same substitutions as for the converse for R . The common randomness rate, distortion, and cardinality bounds follow in the same way as in Section IV-B and by using the support lemma. ■

VII. EXAMPLE

To illustrate the trade-off between communication rate, common randomness rate, and distortion, we provide a simplified achievable rate region from Theorem 2 by assuming that the side information is empty and the distortion measure is the Hamming distortion. More specifically, we consider $X \sim \text{Bern}(0.5)$ and $Z = \emptyset$, with a binary symmetric channel (BSC) from uniformly-distributed U to X with crossover probability α and a BSC from U to Y with crossover probability β , respectively. The resulting achievable rate region

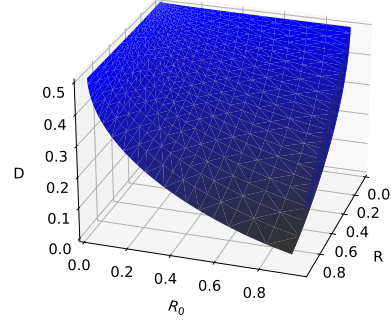


Fig. 3. The boundary of an achievable rate region for the communication rate R , common randomness rate R_0 , and distortion D with strong secrecy and realism defined in Definition 1. For distortions > 0.5 , the achievable region extends along the distortion (vertical) axis beyond $D = 0.5$ for all $R \geq 0$ and $R_0 \geq 0$.

is given as an inner bound on the exact rate region $\mathcal{R}_{\text{BSC},B}$, equal to the rate region given in [16, Theorem 1] given that the side-information Z^n is empty in this case. The proof sketch for the achievability proof of Corollary 2 is given below.

For $\alpha, \beta \in [0, 1]$, define

$$\alpha * \beta = \alpha(1 - \beta) + \beta(1 - \alpha). \quad (38)$$

Denote the binary entropy function as $H_b(p) = -p \log(p) - (1 - p) \log(1 - p)$ for $p \in [0, 1]$.

Corollary 2: The region $\mathcal{R}_{\text{BSC},B}$ includes the union over all joint probability distributions $P_{XUY} = Q_X P_{UY|X}$ of the set of all (R, R_0, D) tuples and $\alpha, \beta \in [0, 1]$ that satisfy

$$R \geq 1 - H_b(\alpha), \quad (39)$$

$$R_0 \geq 1 - H_b(\beta), \quad (40)$$

$$D \geq \alpha * \beta \quad (41)$$

such that $X - U - Y$ forms a Markov chain.

Proof sketch: With $X \sim \text{Bern}(0.5)$, $Z = \emptyset$, and $U \sim \text{Bern}(0.5)$, (X, Z, U, Y) satisfies the conditions in (1)-(3) and the Markov chain $X - U - Y$. The rates in Theorem 1 evaluate to

$$R \geq I(U; X) = 1 - H_b(\alpha), \quad (42)$$

$$R_0 \geq I(U; Y) = 1 - H_b(\beta) \quad (43)$$

since the sum-rate constraint becomes inactive without the side information Z . With Hamming distortion, we have

$$D \geq \mathbb{E}[d(X, Y)] = \sum_{x, y \in \{0, 1\}} p_{XY}(x, y) \mathbb{1}\{x \neq y\} = \alpha * \beta \quad (44)$$

where $\mathbb{1}\{\cdot\}$ is the indicator function. ■

The boundary region of the achievable region in Corollary 2 is depicted in Fig. 3, which illustrates that for a given communication rate, the distortion decreases as the common randomness rate increases. Similarly, for a given common randomness rate, the distortion also decreases as the communication rate increases. Thus, there is a strict trade-off between the two rates, similar to [13, Fig. 2], although there is no explicit sum-rate constraint in the rate region. For

instance, at distortion values close to 0.1 and in the small common randomness rate regime, the communication rate can be reduced by 45 – 52% with an increase in common randomness by 40 – 87%. Similarly, at distortion values close to 0.4 and when the common randomness rate is comparable to the communication rate, the communication rate can be reduced by approximately 31 – 39% with an increase in common randomness by 43 – 63%. These results illustrate the substantial communication rate gains enabled by common randomness in RDP problems, which cannot be achieved in RD problems.

VIII. CONCLUSION

We characterized a tight region for secure RDP with side information when it is available at both the encoder and the decoder, as well as an inner bound for when the side information only is available at the decoder. The latter was shown to be tight for the special case when the side information and the output are jointly i.i.d. We also provided an example to illustrate the remarkable communication rate gains due to common randomness. These remarkable gains are consistent with the general behavior of RDFC problems, which encompass RDP as a special case.

ACKNOWLEDGMENTS

This work was partially supported by the ZENITH Research & Leadership Fund, Swedish Foundation for Strategic Research (SSF), and German Federal Ministry of Research, Technology and Space (BMFTR) 6GEM+ Research Hub.

REFERENCES

- [1] D. Gündüz *et al.*, “Timely and massive communication in 6G: Pragmatics, learning, and inference,” *IEEE BITS Inf. Theory Mag.*, Oct. 2023.
- [2] —, “Beyond transmitting bits: Context, semantics, and task-oriented communications,” *IEEE J. Sel. Areas Commun. (JSAC)*, vol. 41, no. 1, pp. 5–41, Nov. 2022.
- [3] R. Dobrushin and B. Tsybakov, “Information transmission with additional noise,” *IRE Trans. Inf. Theory (T-IT)*, vol. 8, no. 5, pp. 293–304, Sep. 1962.
- [4] T. Berger, “Rate-distortion theory,” *Wiley Encycl. Telecommun.*, 2003.
- [5] I. Csiszár and J. Körner, *Information Theory: Coding Theorems for Discrete Memoryless Systems*. Cambridge University Press, 2011.
- [6] O. Günlü, “Randomized distributed function computation with semantic communications: Applications to privacy,” in *IEEE Int. Workshop Inf. Forensics Security (WIFS)*, Dec. 2024, pp. 1–6.
- [7] M. Hegazy *et al.*, “Compression with exact error distribution for federated learning,” in *Int. Conf. Artificial Intel. Statistics (AISTATS)*, May 2024, pp. 613–621.
- [8] O. Günlü, “Randomized distributed function computation (RDFC): Ultra-efficient semantic communication applications to privacy,” Cryptology ePrint Archive, Paper 2025/1370, 2025. [Online]. Available: <https://eprint.iacr.org/2025/1370>
- [9] A. Shah *et al.*, “Optimal compression of locally differentially private mechanisms,” in *Int. Conf. Artificial Intel. Statistics (AISTATS)*, Mar. 2022, pp. 7680–7723.
- [10] G. Flamich, M. Havasi, and J. M. Hernández-Lobato, “Compressing images by encoding their latent representations with relative entropy coding,” *Adv. Neural Inf. Process. Sys. (NeurIPS)*, vol. 33, pp. 16 131–16 141, Dec. 2020.
- [11] M. Havasi, R. Peharz, and J. M. Hernández-Lobato, “Minimal random code learning: Getting bits back from compressed model parameters,” in *Int. Conf. Learn. Representations (ICLR)*, May 2019.
- [12] B. Isik *et al.*, “Adaptive compression in federated learning via side information,” in *Int. Conf. Artificial Intel. Statistics (AISTATS)*, vol. 238, May 2024, pp. 487–495.
- [13] P. Cuff, “Distributed channel synthesis,” *IEEE Trans. Inf. Theory*, vol. 59, no. 11, pp. 7071–7096, Aug. 2013.
- [14] O. Günlü, M. Skorski, and H. V. Poor, “Low-latency rate-distortion-perception trade-off: A randomized distributed function computation application,” in *EuCNC & 6G Summit*, June 2025.
- [15] D. Bergström and O. Günlü, “Deep randomized distributed function computation (DeepRDFC): Neural distributed channel simulation,” in *IEEE Int. Symp. Inf. Theory (ISIT)*, June 2025.
- [16] G. Åhlgren and O. Günlü, “Secure rate-distortion-perception trade-off over channels: A randomized distributed function computation (RDFC) application,” in *IEEE Int. Symp. Inf. Theory (ISIT)*, Michigan, June 2025.
- [17] K. Sayood, *Introduction to Data Compression*, 4th ed. Morgan Kaufmann, 2017.
- [18] Y. Blau and T. Michaeli, “Rethinking lossy compression: The rate-distortion-perception tradeoff,” in *Int. Conf. Mach. Learn. (ICML)*, June 2019, pp. 675–685.
- [19] R. Matsumoto, “Introducing the perception-distortion tradeoff into the rate-distortion theory of general information sources,” *IEICE Commun. Express (ComEX)*, vol. 7, no. 11, pp. 427–431, 2018.
- [20] J. Chen *et al.*, “On the rate-distortion-perception function,” *IEEE J. Sel. Areas Inf. Theory (JSAIT)*, vol. 3, no. 4, pp. 664–673, Dec. 2022.
- [21] G. Zhang *et al.*, “Universal rate-distortion-perception representations for lossy compression,” *Adv. Neural Inf. Process. Sys. (NeurIPS)*, vol. 34, pp. 11 517–11 529, Dec. 2021.
- [22] L. Theis and A. B. Wagner, “A coding theorem for the rate-distortion-perception function,” in *Neural Compression: From Information Theory to Applications—Workshop@ ICLR*, May 2021.
- [23] N. Saldi, T. Linder, and S. Yüksel, “Output constrained lossy source coding with limited common randomness,” *IEEE Trans. Inf. Theory (T-IT)*, vol. 61, no. 9, pp. 4984–4998, Sep. 2015.
- [24] A. B. Wagner, “The rate-distortion-perception tradeoff: The role of common randomness,” *arXiv preprint arXiv:2202.04147*, Feb. 2022.
- [25] Y. Hamdi, A. B. Wagner, and D. Gündüz, “The rate-distortion-perception trade-off: The role of private randomness,” *arXiv preprint arXiv:2404.01111*, Apr. 2024.
- [26] I. Goodfellow *et al.*, “Generative adversarial nets,” *Adv. Neural Inf. Process. Sys. (NeurIPS)*, pp. 2672–2680, Dec. 2014.
- [27] M. Arjovsky, S. Chintala, and L. Bottou, “Wasserstein generative adversarial networks,” in *Int. Conf. Mach. Learning (ICML)*, Aug. 2017, pp. 214–223.
- [28] I. Gulrajani *et al.*, “Improved training of Wasserstein GANs,” *Adv. Neural Inf. Process. Sys. (NeurIPS)*, vol. 30, pp. 5769–5779, Dec. 2017.
- [29] O. Günlü, M. Bloch, and R. F. Schaefer, “Secure multi-function computation with private remote sources,” in *IEEE Int. Symp. Inf. Theory (ISIT)*, July 2021, pp. 1403–1408.
- [30] C. Schieler and P. Cuff, “Rate-distortion theory for secrecy systems,” in *IEEE Int. Symp. Inf. Theory (ISIT)*, July 2013, pp. 2219–2223.
- [31] A. Kalkhoran *et al.*, “Secure Deep-JSCC against multiple eavesdroppers,” in *IEEE Global Commun. Conf. (GLOBECOM)*, Dec. 2023, pp. 3433–3438.
- [32] A. Wyner and J. Ziv, “The rate-distortion function for source coding with side information at the decoder,” *IEEE Trans. Inf. Theory (T-IT)*, vol. 22, no. 1, pp. 1–10, 1976.
- [33] Y. Hamdi, A. B. Wagner, and D. Gündüz, “Rate-distortion-perception trade-off with strong realism constraints: Role of side information and common randomness,” 2025. [Online]. Available: <https://arxiv.org/abs/2507.14825>
- [34] Y. Hamdi and D. Gündüz, “The rate-distortion-perception trade-off with side information,” in *IEEE Int. Symp. on Inf. Theory (ISIT)*, 2023, pp. 1056–1061.
- [35] X. Niu *et al.*, “Conditional rate-distortion-perception trade-off,” in *IEEE Int. Symp. on Inf. Theory (ISIT)*, 2023, pp. 1068–1073.
- [36] M. H. Yassaee, M. R. Aref, and A. Gohari, “Achievability proof via output statistics of random binning,” *IEEE Trans. Inf. Theory (T-IT)*, vol. 60, no. 11, pp. 6760–6786, Nov. 2014.
- [37] G. Cervia, T. J. Oechtering, and M. Skoglund, “Remote joint strong coordination and reliable communication,” in *IEEE Int. Symp. Inf. Theory (ISIT)*, June 2020, pp. 932–937.
- [38] A. E. Gamal and Y.-H. Kim, *Network Information Theory*. Cambridge, U.K.: Cambridge University Press, 2011.